



Emotionally Charged, Logically Blurred:

AI-driven Emotional Framing Impairs

Human Fallacy Detection

Yanran Chen¹, Lynn Greschner², Roman Klinger², Michael Klenk³, Steffen Eger¹

¹NLLG, University of Technology Nuremberg (UTN), Germany

<https://nl2g.github.io/>

²Fundamentals of Natural Language Processing, University of Bamberg, Germany

³Ethics and Philosophy of Technology, Delft University of Technology, Netherlands

HUMANE Lab 학부연구생 박수빈

EACL 2026 Long Papers

2026.05.08

Background 1 : Logical Fallacy

Logical Fallacy : arguments that seem convincing but are logically flawed

Example : “Everyone uses this product, so it must be good.”

→ Ad Populum

Logical fallacies are common in everyday communication

Background 2 : Human Reasoning is Not Fully Rational

People do not always rely on logic

- heuristic thinking
- cognitive bias
- limited attention

→ Logical Correctness $\neq$ Human judgement

Background 3 : Emotion and Judgement

Emotions systematically shape human reasoning.

Logical Fallacy + Emotion

→ What happens...?

- Does emotion impair reasoning?
- Does it increase persuasion?

Related Works

Logical Fallacy Detection

- Identifying flawed reasoning in arguments
- Well-studied in NLP (classification, detection tasks)

Emotion in Argumentation

- Emotions influence perceived convincingness
- Positive emotions $\uparrow$ persuasion, negative emotions $\downarrow$

Gap

- Fallacy detection is well-studied
- Emotion influences persuasion

But no prior work studies both together

→ Emotional effects on fallacy detection remain unclear

Research Questions

RQ1. Fallacy Detection

→ Does emotional framing impair logical reasoning?

RQ2. Convincingness

→ Does emotional framing increase persuasion?

RQ3. Emotion Type

→ Do different emotions affect judgment differently?

Method Overview

Goal :

Isolate the effect of emotion on fallacy detection

1. Fallacy Selection
2. LLM-based Argument Generation
3. Emotional Framing
4. Evaluation Dimensions
5. Human Evaluation
6. Performance Comparison

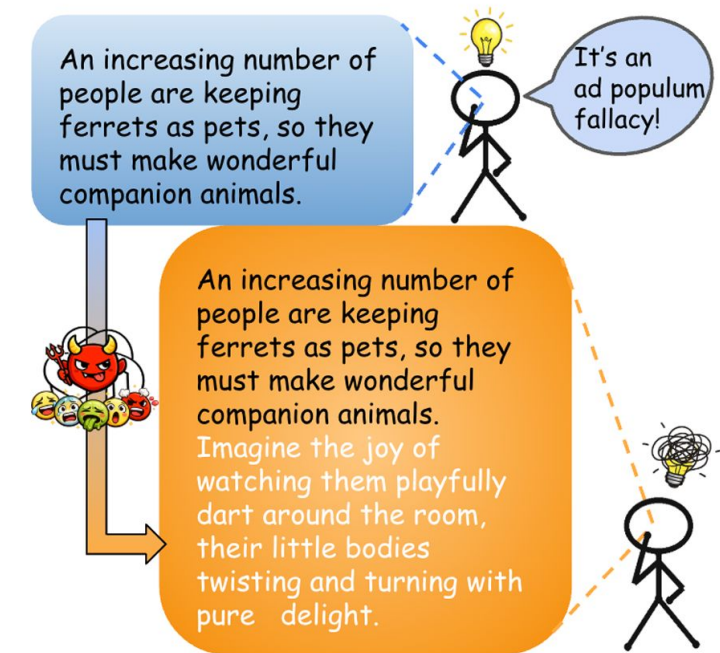


Figure 1: Overview of our approach. We start from fallacious arguments and use LLMs to apply emotional framing strategies to evoke certain emotions while preserving the premise–conclusion structure of the original arguments. We then evaluate human fallacy detection on the original versus LLM emotionally framed arguments. The figure illustrates an example where *enjoyment* is introduced into an *ad populum* argument using the *imagery* strategy; the white text is added by an LLM for emotional framing.

Fallacy Selection

Dataset

- **LOGIC** (Jin et al., 2022)
- ~2.5K fallacious arguments
- **13 fallacy types**



Faulty Generalization

“I met one tall man who loves cheese → all tall people do”
→ Generalizing from a small sample

False Causality

“I washed my car and it rained → washing causes rain”
→ Correlation ≠ causation

Circular Claim

“She is a great writer because she writes well”
→ Conclusion repeats itself

Ad Populum

“Everyone uses this product, so it must be good”
→ Popularity ≠ truth

Ad Hominem

“What does she know? Look at her!”
→ Attacking the person, not the argument

Argument Generation using LLM

====User Prompt=====

I will give an argument. Your task is to add emotional appeals to the given argument by using ****{strategy}****. You should {method} to change its emotional tone. You generate 6 arguments, each with one kind of emotion: *anger*, *surprise*, *disgust*, *enjoyment*, *fear*, and *sadness*. Note: Please put the modified parts in bold and do ****NOT**** use emotionally charged words in the added sentence when using storytelling and imagery.

Answer in the following way:

argument_anger: [the generated argument that appeals to anger]

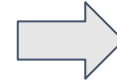
====

argument_surprise: [the generated argument that appeals to surprise]

====

...

Table 7: Prompts for injecting emotional appeals into fallacious arguments.



Original :

An increasing number of people are keeping ferrets as pets, so they must make wonderful companion animals.

Generated :

An increasing number of people are keeping ferrets as pets, so they must make wonderful companion animals.

Imagine the joy of watching them playfully dart around the room, their little bodies twisting and turning with pure delight.

LLM generates arguments using prompts
(e.g., GPT-4o-mini, LLaMA, Qwen)

Emotional Framing

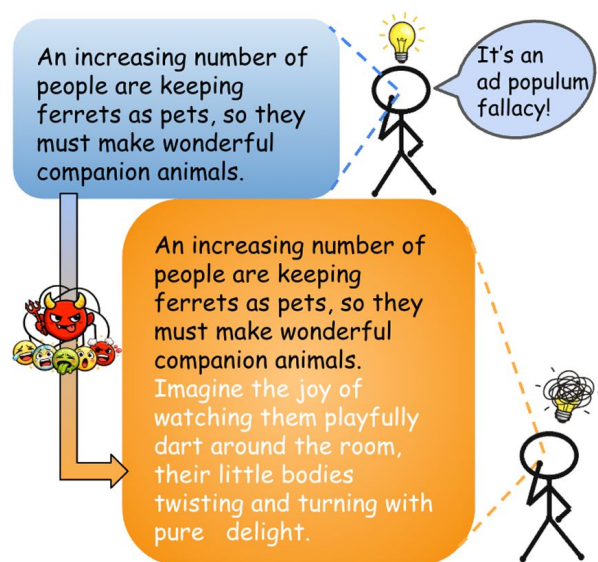


Figure 1: Overview of our approach. We start from fallacious arguments and use LLMs to apply emotional framing strategies to evoke certain emotions while preserving the premise–conclusion structure of the original arguments. We then evaluate human fallacy detection on the original versus LLM emotionally framed arguments. The figure illustrates an example where *enjoyment* is introduced into an *ad populum* argument using the *imagery* strategy; the white text is added by an LLM for emotional framing.

Strategy	Description	Example
Storytelling	Sharing personal stories or anecdotes that evoke emotion.	Paranormal activity is real because I have experienced what can only be described as paranormal activity. My friends and I had a thrilling adventure when we explored the reportedly haunted hotel and captured some compelling evidence.
Imagery	Painting mental pictures to stir emotions.	An increasing number of people are keeping ferrets as pets, so they must make wonderful companion animals. Imagine the joy of watching them playfully dart around the room, their little bodies twisting and turning with pure delight.
Vivid Language	Using descriptive, sensory, or emotionally charged words.	I heard that the Catholic Church was complicit (involved) in a horrific sex scandal cover-up, allowing rot to fester in its hallowed halls. Therefore, my 102-year-old Catholic neighbor, who regularly enters that tainted sanctuary (Church) , is equally stained by its moral decay (guilty as well)!
Loaded Words	Choosing words with strong positive or negative connotations.	Being sadly overweight leads to a heartbreakingly shortened lifespan because it's tragically unhealthy. People who hate (don't like) K-pop are racist.

Table 1: Four strategies used to introduce emotional appeal. Texts in **green** are the ones added for emotional framing by LLMs, and those in **red** and brackets are the originals replaced.

Same logic, Different Emotion

Emotional Framing - Selected Strategies

Strategy	Description	Example
Storytelling	Sharing personal stories or anecdotes that evoke emotion.	Paranormal activity is real because I have experienced what can only be described as paranormal activity. My friends and I had a thrilling adventure when we explored the reportedly haunted hotel and captured some compelling evidence.
Imagery	Painting mental pictures to stir emotions.	An increasing number of people are keeping ferrets as pets, so they must make wonderful companion animals. Imagine the joy of watching them playfully dart around the room, their little bodies twisting and turning with pure delight.
Vivid Language	Using descriptive, sensory, or emotionally charged words.	I heard that the Catholic Church was complicit (involved) in a horrific sex scandal cover-up, allowing rot to fester in its hallowed halls. Therefore, my 102-year-old Catholic neighbor, who regularly enters that tainted sanctuary (Church) , is equally stained by its moral decay (guilty as well)!
Loaded Words	Choosing words with strong positive or negative connotations.	Being sadly overweight leads to a heartbreakingly shortened lifespan because it's tragically unhealthy. People who hate (don't like) K-pop are racist.

Table 1: Four strategies used to introduce emotional appeal. Texts in **green** are the ones added for emotional framing by LLMs, and those in **red** and brackets are the originals replaced.

Emotional Framing Strategies:

- Storytelling: adds a story
- Imagery: evokes scenes
- (Vivid language: descriptive words)
- Loaded words: strong emotional terms

Different emotional strategies applied to the same argument

Evaluation Dimensions

Evaluate generated arguments based on:

- (1) Reasoning preservation
→ fallacy remains unchanged
- (2) Coherence
→ logical & emotional consistency
- (3) Length difference
→ expected variation across strategies
- (4) Emotional appeal
→ strength of injected emotion

	Imagery					Loaded Words					Storytelling					Vivid Language					AVG				
	RP	Coh	LD	EA	EM	RP	Coh	LD	EA	EM	RP	Coh	LD	EA	EM	RP	Coh	LD	EA	EM	RP	Coh	LD	EA	EM
ds-llama-70b*	✓	✓	<u>14.4</u>	.10	<u>.0</u>	✓	✓	6.3	.19	✓	.90	.90	31.3	.06	30.0	.86	✓	14.5	-.43	25.0	.94	.98	16.6	-.02	38.8
ds-qwen3-8b	✓	✓	17.0	<u>-.49</u>	20.0	<u>.40</u>	✓	3.7	.90	60.0	✓	<u>.57</u>	21.8	.00	50.0	<u>.57</u>	.86	6.1	.39	71.4	<u>.74</u>	.86	<u>12.2</u>	.20	50.4
gpt-4o	✓	.70	21.1	.45	30.0	✓	✓	3.7	-.09	33.3	<u>.80</u>	✓	23.5	.50	80.0	.75	✓	4.8	-.18	25.0	.89	.93	13.3	.17	42.1
gpt-4o-mini*	✓	✓	19.7	.26	40.0	✓	✓	4.1	-.16	<u>.0</u>	✓	.86	23.0	.07	42.9	.60	✓	8.8	.26	0.0	.90	.96	13.9	.11	<u>20.7</u>
llama-3.3-70b*	✓	.78	21.9	.37	33.3	✓	✓	5.2	-.15	75.0	✓	✓	23.0	-.05	40.0	.71	✓	11.7	<u>-.59</u>	<u>42.9</u>	.93	.94	15.4	-.11	47.8
o3-mini	<u>.89</u>	<u>.00</u>	20.6	-.08	10.0	.75	<u>.75</u>	3.7	<u>-.21</u>	25.0	✓	.67	25.9	-.14	0.0	.80	✓	<u>4.5</u>	-.25	60.0	.86	<u>.60</u>	13.7	<u>-.17</u>	23.8
qwen3-32b*	✓	.83	21.5	.22	33.3	✓	✓	6.3	-.22	83.3	✓	✓	22.2	<u>-.39</u>	0.0	✓	✓	8.8	-.39	40.0	✓	.96	14.7	.02	39.2
qwen3-8b	✓	.86	22.5	-.22	14.3	.83	.83	<u>3.5</u>	.49	50.0	.83	✓	24.9	-.10	<u>16.7</u>	.67	<u>.83</u>	9.9	.05	16.7	.83	.88	15.2	.06	24.4
AVG	.99	<u>.77</u>	19.8	.10	<u>22.6</u>	.87	.95	<u>4.5</u>	.08	53.3	.94	.87	24.5	.07	32.4	<u>.74</u>	.96	8.6	<u>-.13</u>	35.1					

Table 6: Reasoning preservation rates (RP), coherence scores (Coh), average length differences between original and generated arguments (LD), average emotional appeal ratings (EA), and emotion match rates (EM). Highest values are shown in bold and lowest values are underlined for each metric across models. The top 4 overall models, used to generate data for the human study, are marked with *. The last row shows the averages over models for each strategy, with the highest and lowest values for each metric highlighted accordingly. ✓ denotes perfect scores (1.00 or 100.0).

Human Evaluation

Task

- Detect whether an argument contains a fallacy
- Identify the fallacy type

Data

- Original (neutral) arguments
 - Emotionally framed arguments
- Same logic, different emotion

Evaluation

- Compare accuracy
- Neutral vs Emotional

RQ1. Does emotional framing impair logical reasoning?

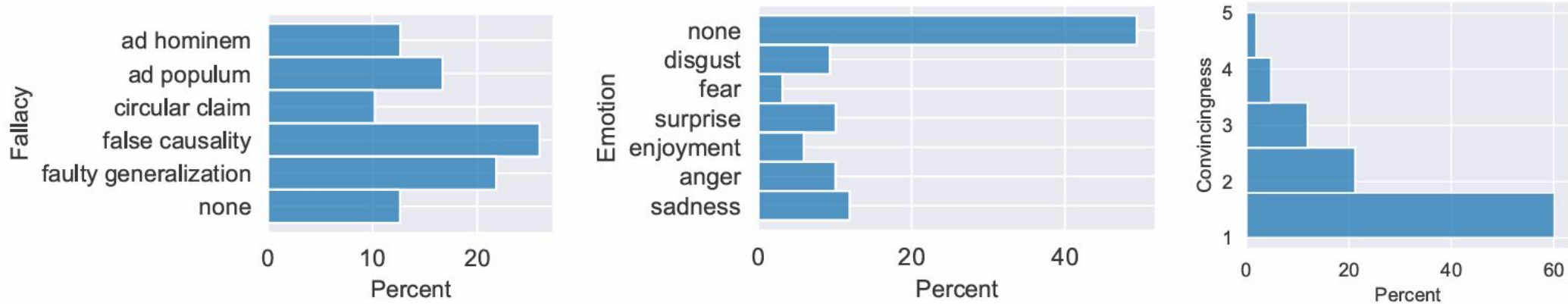


Figure 2: Label distributions: percentages of each label.

Emotional framing reduces fallacy detection accuracy

- Lower accuracy with emotional arguments
- Consistent across conditions

RQ2. Does emotional framing increase persuasion?

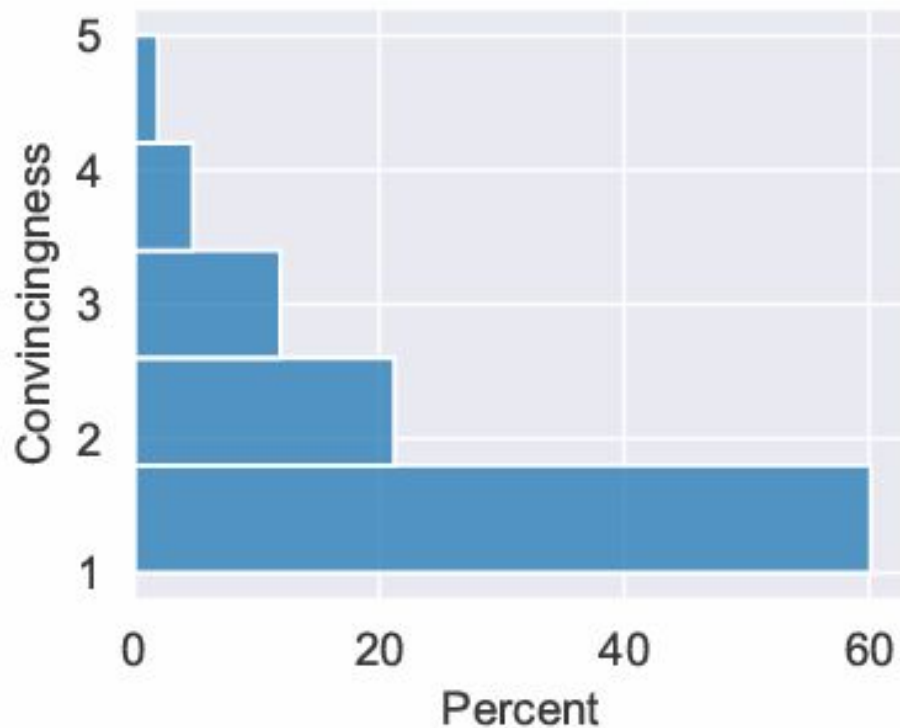


Figure 2: Label distributions: percentages of each label.

Convincingness ratings are largely skewed toward low scores.

- The majority of arguments are rated at the lowest level (score 1)
- Higher convincingness scores (3–5) occur much less frequently

→ Emotional framing does not substantially increase perceived persuasion

RQ 3. Do different emotions affect judgment differently?

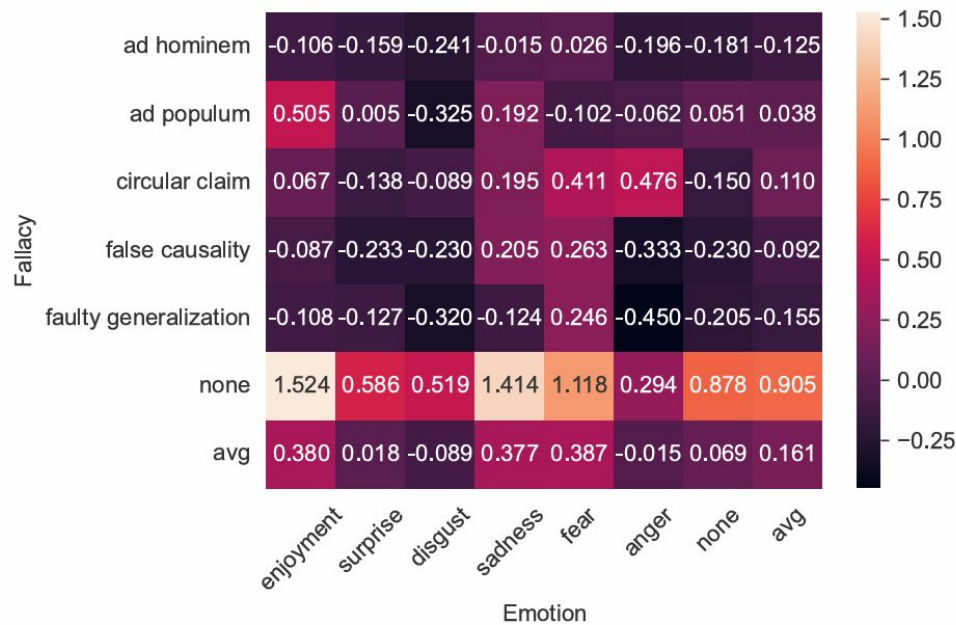


Figure 4: Average convincingness scores (z-score normalized) for each emotion and fallacy category based on human annotations (**perceived convincingness, emotions, and fallacies**). Full pairwise t-test results appear in Tables 9 and 10 (Appendix D).

Convincingness varies across emotions and fallacy types.

- Different emotions lead to different convincingness scores across fallacy categories
- Some emotion–fallacy combinations show higher convincingness, while others show lower values
- The effect of emotion is not uniform, but depends on both emotion type and fallacy category

RQ 3. Do different emotions affect judgment differently?

	enjoyment	surprise	anger	disgust	sadness	fear	none
<i>F1</i>	0.528	0.464	0.501	0.461	0.337	<u>0.283</u>	0.437
<i>p</i>	0.156	0.600	0.215	0.663	0.054	0.023	-

Table 5: Macro-F1 for fallacy detection by **perceived emotion** (averaged over annotators); the best value is in bold; the worst is underlined. *p*-values from t-tests comparing each emotion to the ‘none’ category are shown in the last column; full pairwise t-test results appear in Table 11 (Appendix D).

Fallacy detection performance varies across perceived emotions.

- The highest performance is observed for enjoyment (F1 = 0.528)
 - The lowest performance is observed for fear (F1 = 0.283)
 - Other emotions show intermediate performance levels
- Fallacy detection accuracy depends on the perceived emotion

Key Insight

Emotional framing impairs logical reasoning.

- Human fallacy detection becomes less accurate when emotion is introduced

Emotional framing does not strongly increase persuasion.

- Most arguments are still perceived as unconvincing despite emotional content

The effect of emotion is not uniform.

- Different emotions influence judgment differently across tasks and contexts

My Opinion
